

EFFIPED / TECHNICAL REPORT

Multi-Camera Pedestrian Detection, Tracking & Re-Identification using Joint ConvNeXt V2 Architecture

A compact joint system for pedestrian detection, local tracking, part-aware appearance description, cross-camera candidate retrieval, and analyst review.

62.8%

P-DESTRE validation Rank-1

Cross-camera · Protocol D

64.08

MOT17 val-half MOTA

Protocol A · BoT-SORT

7.78M

Canonical Tier-1

approx. 18 full-pipeline FPS

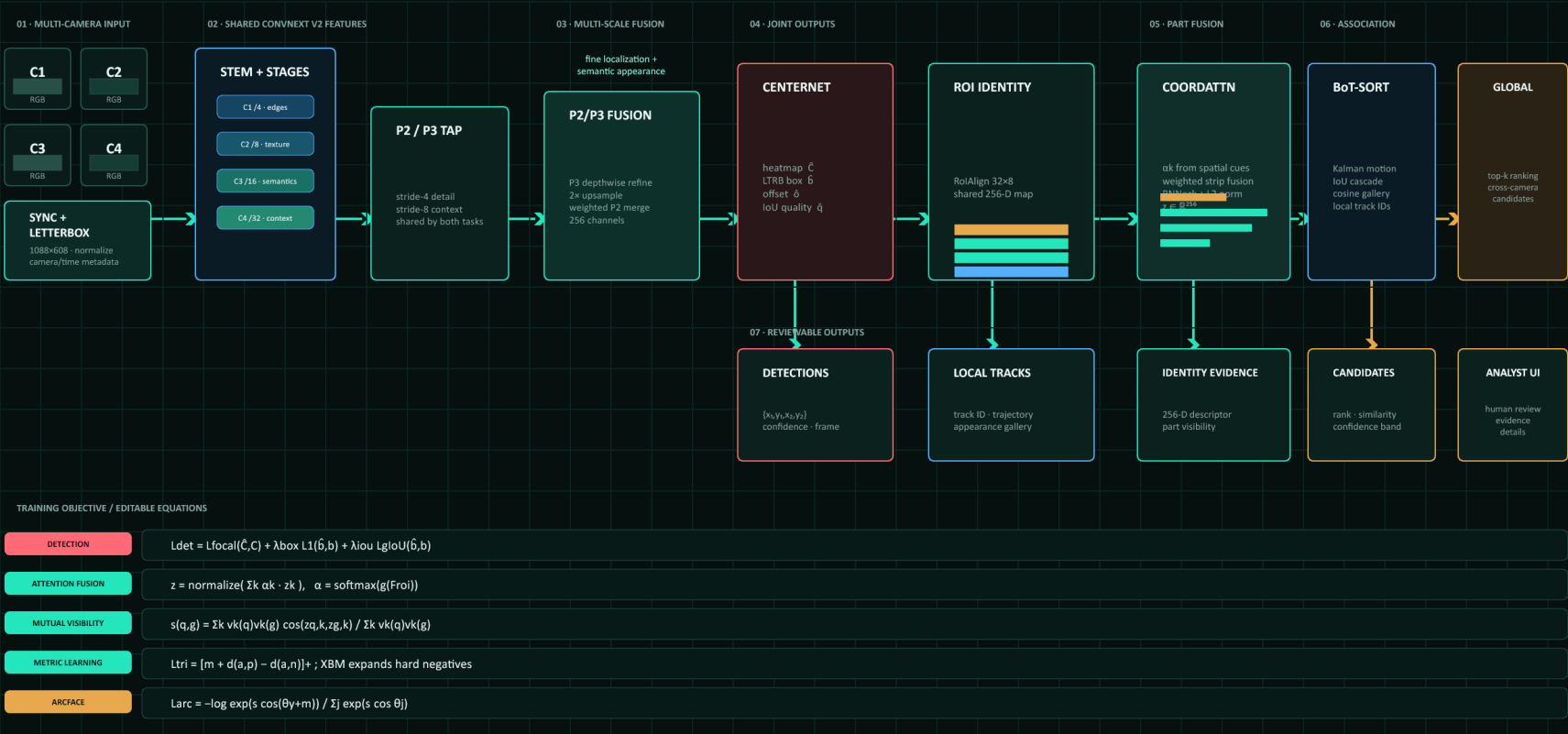
Aswanth Raj · Guide: Sri Preethaa KR · Vertical 1: AI & Intelligent Systems · VIT Vellore School of Computer Science and Engineering (SCOPE)

One visual backbone carries a person from pixels to reviewable evidence.

EffiPed / SYSTEM ARCHITECTURE

3RD PRIZE - SIPC 2026

One shared visual backbone carries a person from pixels to reviewable cross-camera evidence.



DECISION SUPPORT, NOT IDENTITY PROOF

Final record: boxes + local tracks + global candidate identities + evidence-detail view

Aswanth Raj · EffiPed · 2026

The PowerPoint source is fully editable. Detection, four-strip RoI identity readout, CoordinateAttention fusion, BoT-SORT association, and the final analyst evidence view are shown as one connected pipeline.

Contest evidence is separated from the submitted poster and later research.

Evaluation boundary	Rank-1	Detection	Tracking
P-DESTRE fold-0 validation · Protocol D	62.8%	90.74% mAP@0.5	—
P-DESTRE fold-0 test · Protocol E	61.3%	88.4% mAP@0.5	—
MOT17 val-half · Protocol A	—	—	64.08 MOTA · 74.24 IDF1 · 61.34 HOTA

Submitted poster snapshot

The contest poster displayed **7.92M / 22 FPS / 62.8% Rank-1**. The later canonical registry associates Tier-1 with **7.78M parameters and approximately 18 FPS** for the full pipeline. The +16.2 pp row combined multiple changes and is not a pure part-only ablation.

Canonical footprint

Parameters	Input	Descriptor	Benchmark device	Throughput boundary
7.78M	1088×608	256-D	NVIDIA RTX 5080 Laptop GPU	approx. 18 FPS · full pipeline

Every value above is loaded from research/results/summary.json. Unsupported headline claims such as "state of the art," unconditional "real-time," or privacy guarantees are excluded.

The contest prototype became a controlled descriptor-readout investigation.

Stage	Research question	Evidence
EffiPed · contest	Can one compact model detect, track, and describe people across four cameras?	62.8% validation · 61.3% test Rank-1
PartJDE · matched refinement	What is the matched gain from part-aware RoI-strip readout?	+6.66 pp validation Rank-1 · 7.92M · 27.0 FPS
BoxJDE · five-fold	How much does full-box descriptor support change retrieval under a matched JDE model?	+13.31 pp predicted-box Rank-1 · +13.01 pp E2E

Part-aware descriptor

Component	Role	Editable equation
Four body strips	Preserve localized appearance under partial occlusion	$z = \text{normalize}(\sum_k \alpha_k z_k)$
Mutual visibility	Compare only evidence visible in both observations	$s(q,g) = \sum_k v_k(q)v_k(g) \cos(zq,k,zg,k) / \sum_k v_k(q)v_k(g)$
Metric learning	Separate hard identities with XBM-expanded negatives	$L_{tri} = \max(0, m + d(a,p) - d(a,n))$
ArcFace	Angular identity classification during training	$\cos(\theta_y + m)$

Protocol boundary: BoxJDE’s primary P-DESTRE study is a constructed per-date readout ablation, not official Task 4. The public BoxJDE repository carries its full five-fold report rather than duplicating that code here.

A research model is only useful when its uncertainty and data lifecycle remain visible.

Surface	What ships	Safety boundary
Vercel demo	Precomputed four-camera replay with clickable tracks and candidate bands	No upload or remote inference
Local FastAPI mode	Uploads, indexing progress, WebSockets, gallery review, search-by-example	Loopback by default; explicit job deletion
Model artifacts	Versioned manifest and unavailable-weight behavior	Weights withheld pending source-by-source rights review
Media	Optimized still and seven-second VP9 excerpt	P-DESTRE attribution · CC BY-NC-SA 4.0 · non-commercial

Known limitations

Occlusion, pose, lighting, clothing ambiguity, localization error, crowd density, timing, and domain shift can change rankings. A high score is not identity proof.

Responsible operation

Use authorized video, minimize retention, restrict access, preserve human review, validate performance locally, and perform a purpose-specific legal and privacy assessment.

Attribution and links

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